Clara AWS Cost Optimization Executive Report

1. Executive Summary

Clara is operating a multi-account AWS environment supporting a microservices architecture with a current monthly cloud spend of **$250,000 USD**. The highest cost drivers are **EC2, EBS, RDS, S3, EKS, and Lambda**, with EC2 alone representing **40% of total monthly spend**.

The analysis identified several optimization opportunities that can reduce monthly AWS costs without compromising performance, scalability, or operational resilience. The main findings are related to underutilized compute resources, unattached EBS volumes, excessive snapshot retention, high S3 Standard storage usage, and static EKS node capacity.

Based on the current data, the proposed optimization plan can deliver an estimated monthly savings of approximately **$70,000 USD**, equivalent to a **28% reduction** in total AWS spend. The optimized monthly spend would decrease from **$250,000 USD to approximately $180,000 USD**.

This recommendation follows a conservative FinOps approach: first improve visibility and governance, then remove waste, then resize infrastructure, and finally apply commitment-based discounts only after usage patterns are validated.

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2. Current AWS Cost Analysis

Clara's current monthly AWS spend is distributed as follows:

Service	Monthly Cost	% of Total Spend	Main Observations
EC2	$100,000	40%	500 instances running; 60% are c5.xlarge; average CPU utilization is 30%.
EBS	$50,000	20%	200 unattached EBS volumes and excessive snapshot retention.
RDS	$37,500	15%	Opportunity for reserved capacity and storage review.
S3	$25,000	10%	50 TB stored; 80% remains in S3 Standard.
EKS	$25,000	10%	3 clusters with 10 c5.xlarge nodes each; static node capacity.
Lambda	$12,500	5%	Smaller optimization opportunity; review memory and execution duration.
Total	**$250,000**	**100%**	Current baseline before optimization.

The most relevant cost issue is not one single service, but the absence of a mature cost governance model across compute, storage, Kubernetes, and data lifecycle management. The environment shows signs of rapid growth without enough automation around rightsizing, cleanup, tagging, and lifecycle policies.

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3. Prioritized Optimization Recommendations

Recommendation 1: EC2 and EKS Compute Optimization

Priority: High
Estimated Monthly Savings: $42,500 USD
Estimated Impact: 17% of total AWS spend

EC2 and EKS are the largest optimization areas. Clara currently operates 500 EC2 instances, with 60% using c5.xlarge and average CPU utilization of only 30%. This suggests that part of the fleet may be oversized or running continuously without matching workload demand.

Recommended actions:

- Use AWS Compute Optimizer and CloudWatch metrics to identify underutilized EC2 instances.
- Rightsize oversized instances based on CPU, memory, network, and workload behavior.
- Apply Auto Scaling policies where workloads are elastic.
- Use Savings Plans for stable baseline compute usage after rightsizing.
- Introduce Spot Instances only for non-critical, fault-tolerant workloads.
- Optimize EKS node groups using Cluster Autoscaler or Karpenter.
- Use mixed instance types for EKS worker nodes where appropriate.

This should be implemented carefully, beginning with development and staging workloads before production changes.

Recommendation 2: EBS Volume and Snapshot Optimization

Priority: High
Estimated Monthly Savings: $17,500 USD
Estimated Impact: 7% of total AWS spend

EBS represents 20% of Clara's monthly AWS cost. The environment has 200 unattached EBS volumes and approximately 10 snapshots per in-use volume. This indicates direct storage waste and lack of lifecycle governance.

Recommended actions:

- Identify and delete unattached EBS volumes after owner validation.
- Create automated cleanup policies for orphaned storage resources.
- Review EBS volume types and migrate suitable workloads to gp3.
- Implement lifecycle policies for EBS snapshots.
- Define snapshot retention by workload criticality and compliance requirements.
- Apply tagging standards to volumes and snapshots for ownership and cost allocation.

This is a low-risk, high-impact optimization area because unattached volumes and obsolete snapshots usually do not affect active production workloads when properly validated.

Recommendation 3: S3 Storage Lifecycle Optimization

Priority: Medium
Estimated Monthly Savings: $10,000 USD
Estimated Impact: 4% of total AWS spend

Clara stores 50 TB of data in S3, with 80% in the Standard storage class. This suggests that a large portion of storage may not require frequent access and could be moved to lower-cost storage tiers.

Recommended actions:

- Analyze S3 access patterns using S3 Storage Lens.
- Move infrequently accessed objects to S3 Intelligent-Tiering.

- Apply lifecycle policies to transition older data to S3 Standard-IA, Glacier Instant Retrieval, Glacier Flexible Retrieval, or Deep Archive where appropriate.
- Review incomplete multipart uploads and expired objects.
- Apply bucket-level tagging and ownership standards.
- Define retention rules aligned with business and compliance requirements.

This optimization should be applied progressively because storage lifecycle policies can affect retrieval time and cost if not aligned with access requirements.

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4. Estimated Savings Summary

Optimization Area	Current Monthly Cost Area	Estimated Savings	Business Impact
EC2 and EKS Compute Optimization	$125,000	$42,500	Rightsizing, autoscaling, Savings Plans, EKS node optimization.
EBS Optimization	$50,000	$17,500	Remove unattached volumes, optimize volume types, reduce snapshot waste.
S3 Lifecycle Optimization	$25,000	$10,000	Move cold or infrequently accessed data to lower-cost tiers.
Total Estimated Savings	**$200,000 Addressable Spend**	**$70,000**	**Estimated 28% reduction of total AWS monthly spend.**

After implementation, Clara's estimated monthly AWS spend would decrease from:

Metric	Amount
Current Monthly AWS Spend	$250,000
Estimated Monthly Savings	$70,000
Optimized Monthly AWS Spend	$180,000
Estimated Reduction	28%

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5. Implementation Plan

Phase 1: Visibility, Validation, and Governance

Timeline: Weeks 1–2

- Enable or validate AWS Cost Explorer, Cost and Usage Report, AWS Budgets, and Cost Anomaly Detection.
- Confirm tagging standards across accounts, services, applications, environments, and owners.
- Build a cost baseline by service, account, workload, and environment.
- Identify top cost drivers and validate them with engineering teams.
- Define approval rules for deleting or modifying resources.

Phase 2: Waste Removal and Rightsizing

Timeline: Weeks 3–6

- Remove validated unattached EBS volumes.
- Apply EBS snapshot lifecycle policies.
- Review EC2 utilization and rightsize low-utilization instances.
- Optimize EKS node groups and introduce autoscaling.

- Apply S3 lifecycle policies to low-access data.
- Track savings weekly and compare against the original baseline.

Phase 3: Rate Optimization and Continuous FinOps

Timeline: Ongoing

- Apply Savings Plans or Reserved Instances only after usage has been rightsized and stabilized.
- Review RDS reserved capacity opportunities.
- Establish monthly cost reviews with application owners.
- Create dashboards for cost, utilization, savings, and forecast variance.
- Implement governance controls to prevent cost regression.

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6. Risks and Controls

Risk	Mitigation
Rightsizing production workloads too aggressively	Start with non-production, validate performance metrics, and roll out progressively.
Deleting storage still needed by teams	Require owner validation and backup confirmation before deletion.
S3 lifecycle policies affecting retrieval time	Classify data by access pattern and compliance requirement before transition.
Savings Plans purchased before optimization	Rightsize first, then commit only to stable baseline usage.
Cost savings not sustained	Implement budgets, anomaly detection, tagging enforcement, and monthly FinOps reviews.

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7. Conclusion

Clara can reduce its AWS monthly spend by approximately **$70,000 USD**, or **28%**, through a focused optimization plan targeting compute, storage, and lifecycle governance. The recommendations are feasible, measurable, and aligned with a practical FinOps operating model.

The strongest opportunities are EC2/EKS rightsizing, EBS cleanup, snapshot lifecycle management, and S3 storage class optimization. Once these actions are completed, Clara should move into continuous cost governance using Cost Explorer, Cost and Usage Reports, AWS Budgets, Compute Optimizer, and monthly engineering reviews.

This approach reduces cost without weakening performance, scalability, or production reliability.